

Ali Raza

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SUMMARY

AI/ML Engineer with 3+ years building and deploying LLM- and VLM-powered systems in production. Core expertise across RAG pipelines, AI agents, prompt engineering, and multimodal vision-language integration using LangChain, LangGraph, Gemini, and OpenAI/Anthropic APIs. Experienced integrating agentic and voice workflows into full-stack applications via FastAPI with hands-on MLOps for monitoring and deployment.

TECHNICAL SKILLS

ML / DL & CV: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, **Optimization**(LoRA, QLoRA, PEFT) , **ML Models**(XGBoost, LightGBM), fine-tuning, transfer learning, **Computer Vision**(YOLOv8, TransU-Net, U-Net, ViT, SAM, ResNet, DenseNet, BioViL-T, GradCAM, CBAM, OpenCV), **Deep Generative Models**(GANs, VAEs, Probabilistic Models, Diffusion, Mamba) **NLP, LLMs & Data:** BERT, LlamaIndex, LangGraph, RAG pipelines, LangChain, FAISS, Pinecone, embeddings, prompt engineering, OpenAI API, Gemini API, Anthropic API, Pandas, NumPy, SQL, PostgreSQL, MongoDB **MLOps, Deployment & Programming:** MLflow, Docker, Kubernetes, Kubeflow, Prometheus, Grafana, Evidently AI, FastAPI, CI/CD, GitHub Actions, AWS EC2, inference optimization, GPU deployment, Python, C++, JavaScript, Git

EXPERIENCE

ML Engineer — Freelance | International Clients 2023 – Present
Remote, Faisalabad, Pakistan

- Designed **end-to-end ML pipelines** — preprocessing, training, evaluation, and **scalable production deployment** via FastAPI + Docker with real-time **model performance monitoring**.
- Built and deployed **YOLOv8 computer vision system** — inference-optimized REST API, containerized with Docker, GPU deployment on Vast.ai, integrated with React frontend on Vercel.
- Fine-tuned **LLMs (LoRA/QLoRA)** via HuggingFace PEFT; built **RAG pipelines** with LangChain, FAISS, and Pinecone for document intelligence systems.
- Maintained **MLflow** experiment tracking and model versioning; automated deployments via **CI/CD** pipelines using GitHub Actions.

Data Analyst Intern | TAKMIL 2024
Louisville, Kentucky, USA · Remote

- Cleaned and analyzed large-scale datasets via Python (Pandas, NumPy, SQL); built **Power BI dashboards** — proposed optimizations improving workflow efficiency by **30%**.

RESEARCH & KEY PROJECTS

RadGuard: Multimodal ML System for Medical Imaging — Final Year Thesis 2025–2026
BioViL-T · CXR-BERT · PyTorch · MLflow · Docker · Kubernetes · Kubeflow · FastAPI · AWS EC2

- Built multimodal deep learning system detecting 4 error types in AI-generated chest X-ray reports across 14 conditions on 74,060 MIMIC-CXR samples (**Val F1=0.66**); discovered and fixed 10 critical labeling bugs producing 147K false labels.
- Deployed on **AWS EC2** with Docker + Kubernetes autoscaling, **Kubeflow** retraining pipeline, Prometheus+Grafana monitoring, A/B shadow testing, and **FAISS RAG** chatbot for clinical querying.

AI Cost Segregation Platform — Multimodal VLM Pipeline, Production SaaS 2025–Present
Gemini VLM · YOLOE · SAM2 · FastAPI · Firebase · Prompt Engineering

- Technical lead on an AI SaaS converting property photos into audit-ready tax reports.
- Replaced a fine-tuned closed-vocab detector with a **Gemini VLM classification stage** against a 312-item taxonomy — eliminated retraining loop entirely via prompt/taxonomy iteration.
- Designed **3-stage vision pipeline (YOLOE + SAM2 + Gemini)** with IoU deduplication.

ProgressGAN-CXR — Generative AI Research 2025–2026
PyTorch · DCGAN · WGAN-GP · Knowledge Distillation · FastAPI · React · HuggingFace · Docker

- Trained 4 severity-conditioned GAN architectures for continuous COVID-19 disease progression simulation on 21,165 chest X-rays — proposed novel 3-metric evaluation framework (FID, Spearman r, SSIM) showing FID alone is insufficient for medical GANs.
- Deployed full-stack production application — FastAPI inference server on HuggingFace Spaces with Prometheus monitoring, React + TypeScript frontend on Vercel, model weights on HF Hub; live demo at progressgan-cxr.vercel.app.

CV Interview Coach — AI Multi-Agent System 2025
LangGraph · LangChain · Groq · ChromaDB · LlamaIndex · Whisper · LiveKit · FastAPI · Next.js

- Built a **3-agent pipeline** with Supervisor MCP orchestrator — CV/JD analyzer, RAG-augmented question generator, and real-time voice interview agent with **self-healing retry loops** (critique injection, max 3 retries).

EDUCATION

BSc Computer Science | **FAST-NUCES, Faisalabad** 2022 – 2026
Co-Head Hackathon ACM Daira 2025 | TAKMIL Data Science Internship 2024

PUBLICATIONS

Ali Raza et al. “Segmentation & Classification of CIEDs in Chest X-rays” — ACR 2026